

# HARIN: A Novel Metric for Hierarchical Topic Model Assessment

Harin Seo\*

Samsung Electronics Co., Ltd.  
Suwon, Republic of Korea  
haha.seo@samsung.com

Byungchul Tak

Kyungpook National University  
Daegu, Republic of Korea  
bctak@knu.ac.kr

Hwanseong Joo

Kyungpook National University  
Daegu, Republic of Korea  
joohs0505@knu.ac.kr

Young-Kyoon Suh<sup>†</sup>

Kyungpook National University  
Daegu, Republic of Korea  
yksuh@knu.ac.kr

## Abstract

Hierarchical topic modeling is a well-known technique for deriving comprehensive insights on a given dataset. However, it is challenging to choose the best-suited hierarchical topic model among many candidates, given that the model generally depends on the dataset under analysis. Moreover, even that chosen model typically produces an overwhelming number of leaf topics, making it hard to correctly interpret the result derived from the model. Although topic coherence is an often used metric to assess the quality of a model, coherence cannot reflect the unique characteristics of the hierarchical structure when applied to the model as well. To address these concerns, we propose a novel evaluation metric, *HARIN* (Hierarchical haRmony INdex). The proposed HARIN metric effectively reflects the overall topic coherence, diversity, and similarity among parent-child and sibling layers in the produced hierarchy. We test the validity of HARIN by comparing it against competing metrics and human assessments of the quality of topic hierarchies from four popular models applied to five real-world datasets. Our experiments demonstrate that HARIN achieves approximately 85% accuracy in model ranking compared to human scores, surpassing the leading competing metric, coherence, by 1.4X. Notably, HARIN's mean reciprocal rank of 1 highlights its exceptional ability to recommend the optimal hierarchical topic model. We show that HARIN outperforms the coherence metric in identifying the best model while also pruning the right number of leaf topics, thereby enhancing both model selection and result interpretability.

## CCS Concepts

• Information systems → Document topic models; • Applied computing → Document analysis.

## Keywords

Hierarchical Topic Modeling, Hierarchical Harmony Index, Topic Modeling Evaluation

\*Work done at Kyungpook National University

<sup>†</sup>Corresponding author



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## 1 Introduction

Hierarchical topic modeling has been a popular technique to discover significant topics implied in a given dataset and visualize the outcome in a tree-like structure [7]. Much research [1, 13, 15, 28, 29, 42, 43] has used this tool to gain valuable insights from social media datasets. For instance, some past works [13, 28, 29] utilized the hierarchical topic modeling approaches to explore overall thoughts and opinions regarding COVID-19 discourse from Twitter (now, 'X') datasets.

However, several nontrivial concerns arise in employing the hierarchical topic modeling analysis. First, existing hierarchical topic models generate excessive topics, making it almost impossible to interpret their results and capture impactful insights.

For example, a recent hierarchical topic modeling technique [38] produces an overwhelming number of topics in the order of thousands. Second, existing studies [12, 35, 38] typically evaluate the hierarchical topic models only using the coherence metric based on NPMI (Normalized Pointwise Mutual Information) [3]. However, the coherence metric captures only the internal consistency of individual topics (intra-topic relationships) and does *not* account for the relationships between topics (inter-topic). It is designed to assess how semantically similar the words within a single topic are without considering the broader topic structure. Moreover, the coherence score tends to decrease as topics increase for several reasons, like increased complexity of a model, diluted semantic contents, and reduced interpretability. Thus, the coherence metric alone is insufficient as an evaluation tool to determine the optimal number of topics that can yield enhanced interpretability.

To address these concerns, we introduce a novel metric, the *Hierarchical haRmony INdex* (HARIN), to better assess the quality of hierarchical topic models by reflecting the distinct structural characteristics of topics in a holistic fashion. HARIN captures *i)* the *coherence* of the whole topic, *ii)* the *similarity* between parent and children topics, and *iii)* the *diversity* between sibling topics. To evaluate the performance of HARIN, we use four representative hierarchical topic models chosen from each category (embedding-based, classic, and non-probabilistic-based): BERTopic [8], hierarchical Latent

Dirichlet Allocation (hLDA) [7], CluHTM [38], and HyHTM [35]. Through the experiments on five real-world datasets, we demonstrate that the proposed HARIN metric can assist interpreters in choosing the best model among the four models and determining the optimal number of topics at leaf levels. Our results also reveal that hLDA [7] chosen by HARIN is a more effective tool than the others in light of providing diverse and cohesive hierarchical views of the datasets. Using human ratings as the ground truth, HARIN achieves approximately 85% accuracy, offering strong confidence in its ability to select the optimal model without human intervention. This makes it a reliable tool for interpreting hierarchical topic modeling results. Notably, HARIN's mean reciprocal rank [9] of 1 underscores its exceptional performance in consistently recommending the best hierarchical topic model.

The following are the contributions of this paper:

- We introduce HARIN, a novel evaluation metric for hierarchical topic models that fully capture the structural characteristics of a model. HARIN quantifies the topic coherence, the similarity between parent and child topics, and the diversity among sibling topics, providing a holistic view of the model's structure. This approach enables more precise interpretations and insights.
- To validate HARIN, we evaluate four prominent hierarchical topic models across five real-world datasets: one private dataset of Korean Twitter posts mentioning COVID-19 vaccine brands and four public datasets [11, 16, 34, 41]. Our experiments show that hLDA consistently outperforms the alternatives based on our metric.
- We demonstrate that HARIN achieves approximately 85% accuracy when compared to human ratings as the ground truth, with a mean reciprocal rank of 1 across datasets, surpassing the leading competing metric, coherence, by a factor of 1.4.
- We have made our code and datasets publicly accessible on GitHub to ensure reproducibility.

The structure of this paper is as follows. The next section reviews the literature relevant to our study. Section 3 defines the research problem. In Section 4, we introduce the HARIN metric and outline the process for selecting the most suitable hierarchical topic model using HARIN. Section 5 details the experimental results and their analysis. Finally, Section 6 concludes the paper and highlights directions for future research.

## 2 Related Work

**Hierarchical Topic Modeling.** Several traditional techniques, such as Latent Dirichlet Allocation (LDA) [14] and Non-negative Matrix Factorization (NMF) [19], perform flat topic modeling, which has the weakness of being unable to identify the hierarchy of topics implicit in documents. To overcome the shortcomings, researchers have focused on extending conventional topic modeling techniques to obtain hierarchical information on topics.

The following are the three hierarchical topic modeling approaches we consider in this paper.

- *Latent Dirichlet Allocation (LDA) Based Model:* An LDA model represents a document blending multiple topics with distinct words. By combining the probabilities of document-topic and topic-word relationships within a document, LDA simultaneously estimates the distribution of topics within a document

and the distribution of words within a topic. This process determines the extent to which each document relates to a topic and identifies the most prominent words within each topic. However, the drawback of this approach is that the model training and topic generation process can be time-consuming due to its use of Bayesian methods, such as the Gibbs sampler. One of the most renowned traditional models in this category is hierarchical LDA (hLDA), which constructs topic mining and hierarchy based on the concept of nested Chinese Restaurant Process (nCRP) [7]. There have been some other studies extending the hLDA method (e.g., constrained-hLDA [39], KeyETM [10], SSHLDA [22], and thLDA [17]). This paper uses the hLDA model as a basis for comparing other models.

- *Non-Negative Matrix Factorization (NMF) Based Model:* This model relies on NMF, a non-probabilistic technique that decomposes documents and words into non-negative matrices to extract the topic composition of documents and the topic attributes of words. NMF assumes that each document consists of a collection of topics, and each topic comprises a set of words. NMF combines the document-topic and topic-word matrices to approximate the original data. Many researchers have conducted their studies to enhance hierarchical topic modeling by applying NMF to neural networks (e.g., TSNTM [12]) or incorporating semantic representations through word embeddings (e.g., CluHTM [38] and HyHTM [35]).
- *Embedding Based Model:* Recent advancements in advanced language embedding models such as BERT [5] and GPT [4] have led to the exploration of clustering based on embeddings. This approach involves extracting topics from separated clusters and conducting hierarchical topic modeling by constructing a hierarchy based on the extracted topics. (e.g., BERTopic [8], Top2Vec [2], and TaxoCom [20]).

**Topic Modeling Evaluation.** Common evaluation metrics for topic modeling include *topic coherence*, *topic significance*, *topic specialization*, *diversity*, *similarity*, and *classification* metrics [18, 36]. Topic coherence measures the degree of semantic coherence among the top- $k$  words within a topic. Topic significance measures the quality or importance of the identified topics. Topic specialization measures the semantic distance of each topic from a general baseline topic. Diversity assesses how various the top- $k$  words are within each topic. Similarity means the proximity of the top- $k$  words assigned to a topic. Classification assesses the performance of a classifier to predict the document class based on the distribution of topics.

Unfortunately, none of these existing metrics fully captures the inherent hierarchical structure of topics on their own, highlighting the need for a new evaluation metric specifically designed for hierarchical topic modeling. Lindgren [21] proposed a novel evaluation metric based on word frequency to evaluate hierarchical topic modeling; however, it only considers the parent-child relationship. Shahid et al. [35] used HCoherence and Hierarchical Affinity to evaluate the quality of a hierarchical structure. Overall, the existing metrics merely considered vertical and horizontal relationships of a produced topic hierarchy rather than a single harmonized metric.

Meanwhile, this paper introduces a new hierarchical topic model evaluation metric, which holistically reflects in its assessment the

structural characteristics of hierarchical topic modeling: The metric evaluates (i) the similarity between parent and child topics, (ii) the diversity among sibling topics, and (iii) the overall coherence of the topic hierarchy. By integrating these measures, the metric comprehensively assesses the hierarchy's structure. Moreover, it suggests a basis for pruning less impactful leaf topics, which, when done effectively, can further enhance the interpretability of the hierarchy by reducing complexity and focusing on the most meaningful topics. In particular, note that the latter issue has been overlooked in the existing metrics.

### 3 Problem Definition

Our goal is to find the most suitable scoring metric,  $\mathcal{M}$ , that can assess the overall quality of the topic hierarchy produced by a topic model  $l \in \mathcal{L}$ , which is a set of hierarchical topic models in consideration. To unravel the overall discourse implied in a text dataset  $\mathcal{T}$ , we choose the topic hierarchy,  $\mathcal{H}$ , of the model yielding the highest value of  $\mathcal{M}$ , which is defined as a weighted combination of three components—similarity ( $S$ ), diversity ( $D$ ), and coherence ( $C$ ).  $\mathcal{H}$  represents a topic tree structure and is defined as  $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ , where each node ( $\in \mathcal{V}$ ) indicates a group of related keywords to best depict its single conceptual topic, and each edge ( $\in \mathcal{E}$ ) expresses a hierarchical relationship between a parent node ( $\in \mathcal{V}$ ) and its child ( $\in \mathcal{V}$ ). A topic node,  $v_i \in \mathcal{V}$ , is the ordered set of keywords belonging to the same topic based on the degree of relevance to the topic. Each textual document (e.g., a tweet or a news article)  $d_j \in \mathcal{T}$  is expressed as a sequence of terms  $(e_j^1, \dots, e_j^k)$ , where  $e_j^k$  indicates a distinct  $k$ -th term extracted from  $d_j$  in  $\mathcal{T}$ .  $e_j^k$  should be from the legal vocabulary set  $\mathcal{O}$  in a language  $\mathcal{G}$ .  $S$ ,  $D$ , and  $C$  mean how similar the parent and child topic nodes in  $\mathcal{H}$ , how various the sibling nodes under the same parent are, and how close the topic words in the same nodes are overall to each other. The performance of  $\mathcal{M}$  is determined by the proximity to a human-generated rating and by how many leaf topics are left after pruning on  $\mathcal{H}$ .

Our hierarchical topic modeling evaluation policy is to determine the most representative topic hierarchy,  $\mathcal{H}$ , with the highest score of  $\mathcal{M}$ , to replace human assessments. As the values of  $S$ ,  $D$ , and  $C$  increase, the value of  $\mathcal{M}$  also increases.

### 4 Proposed Approach

In this section, we elaborate on our new evaluation metric, *Hierarchical Harmony Index* (HARIN), that integrates the horizontal and vertical structural characteristics in a given topic hierarchy.

#### 4.1 Hierarchical Harmony Index

**Definition of Hierarchical Harmony Index:** Our hierarchical harmony index, HARIN, is formulated as shown in Equation 1:

$$\text{HARIN} = w_d D_H + w_s S_H + w_c C, \quad (1)$$

where  $D_H$  and  $S_H$  are the hierarchy-wide diversity and similarity scores, respectively,  $C$  is a coherence score widely used to evaluate topic models, and  $w_d$ ,  $w_s$ , and  $w_c$  represent the weights of  $D_H$ ,  $S_H$ , and  $C$ , such that  $w_d + w_s + w_c = 1$ .  $D_H$ ,  $S_H$ , and  $C$  are between 0 and 1. Each weight can be configured differently depending on whether researchers wish to emphasize similarity (between parents and children), diversity (between siblings), or coherence (of all topics)

when determining the best hierarchical topic model. We set  $w_i$  to an equal weight among the three component scores. The computed HARIN score in Equation 1 quantifies that the higher the similarity between the parent and its child, the higher the diversity between siblings, and the higher the overall coherence of a hierarchy, the better the quality of a given hierarchical topic model. We calculate  $D_H$ ,  $S_H$ , and  $C$  based on the top 20 words in a given topic that are most significant.

**Similarity Based Word Embedding:** Word embedding is a way to represent words as vectors. We use the Word2Vec [26] model in the gensim package [32] to calculate the significant similarity between words and obtain the similarity between two topics. We derive  $S$  by measuring the similarity of words between parent-child topics as shown in the following Equation 2:

$$S(T_1, T_2) = \frac{\sum_{i=1}^n \sum_{j=1}^m \text{Sim}(T_{1i}, T_{2j})}{n \times m}, \quad (2)$$

where  $T_1$  and  $T_2$  represent two topics, respectively.  $\text{Sim}(T_{1i}, T_{2j})$  calculates the word-embedding-based similarity between  $T_1$ 's  $i$ -th word and  $T_2$ 's  $j$ -th word. We utilize the Word2Vec model trained on the collected data to compute the similarity. If two topics in a parent-child relationship are almost the same, the parent does not have to bear a child. The penalty halves the derived similarity if more than half of the words in two topics between parent and child are the same.

**Topic Diversity based Word Embedding:** This metric ( $D$ ) measures the degree of diversity between two topics. We derive  $D$  based on word embedding-based distance as shown in Equation 3:

$$D(T_1, T_2) = 1 - S(T_1, T_2), \quad (3)$$

where  $T_1$  and  $T_2$  represent two topics, respectively. We derive  $D$  by examining how many various sibling topics exist.

**Topic Coherence:** To consider the quality of the topic as a whole beyond its hierarchical characteristics, we incorporate into HARIN coherence, which is often used as a topic modeling evaluation metric. Normalized PMI (Pointwise Mutual Information) [3] and coherence  $C$  are defined as follows:

$$\text{NPMI}(o_i, o_j) = -\frac{\left(\log \frac{p(o_i, o_j)}{p(o_i)p(o_j)}\right)}{\log p(o_i, o_j)}. \quad (4)$$

$$C(o_i, o_j) = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{NPMI}(o_i, o_j)}{\binom{n}{2}}, \quad (5)$$

where  $n$  is the number of words belonging to the topic, and  $o_i$  and  $o_j$  are the  $i$ -th and  $j$ -th words belonging to the topic, respectively. NPMI is used to assess the semantic coherence of a topic by measuring the strength of association between the words within that topic. Higher NPMI values indicate that the words in a topic are semantically related, contributing to a more coherent and interpretable topic. The coherence value ranges from -1 to 1 and is normalized between 0 and 1, like the similarity and diversity values.

A coherence score is normalized between 0 and 1, as are the scores of similarity and diversity.

**Algorithm 1:** Calculating the Hierarchical Harmony Index (HARIN) Score on a Given Hierarchical Topic Model

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**Input** : A given hierarchical topic model ( $m$ ), the number of top words ( $k$ ) belonging to each node in the topic hierarchy ( $H$ ) produced by  $m$ , and a weight vector  $W = [w_s, w_d, w_c]$

**Output**: Calculated the HARIN score ( $score$ )

```

1 Function CalcDiversity( $node, k$ ):
2    $D \leftarrow 0$ ;
3    $child\_list \leftarrow node.childs$ ;
4   for  $i \leftarrow 0$  to  $len(child\_list) - 1$  do
5     for  $j \leftarrow 0$  to  $i - 1$  do
6        $D += (1 - getSimilarity(child\_list[i], child\_list[j], k))$ ;
7       // Apply Eq. 3.
8     end
9    $D /= \sum_{i=0}^{len(child\_list)-1} i$ ;
10  return  $D$ ;
11 End Function
12 Function CalcSimilarity( $node, k$ ):
13   $S \leftarrow 0$ ;
14  foreach  $c \in node.childs$  do
15     $S += getSimilarity(node, child, k)$ ; // Apply Eq. 2.
16  end
17   $S /= |node.childs|$ ;
18  return  $sim\_s$ ;
19 End Function
20 Function CalcTotalScore( $m, k, W$ ):
21   $H \leftarrow$  Obtain a topic hierarchy produced by  $m$ .
22   $D\_H \leftarrow 0$ ;
23   $S\_H \leftarrow 0$ ;
24   $C \leftarrow H.get\_coherence()$ ;
25  for  $l \leftarrow 0$  to  $H.level$  do
26    foreach  $node \in H.nodes(level)$  do
27      if  $node.childs == NULL$  then continue;
28       $D\_H += CalcDiversity(node, k)$ ;
29       $S\_H += CalcSimilarity(node, k)$ ;
30    end
31  end
32   $D\_H /= |H.nodes| - |H.leaf|$ ;
33   $S\_H /= |H.nodes| - |H.leaf|$ ;
34  return  $(w_d \times D\_H + w_s \times S\_H + w_c \times C)$ ; // Apply Eq. 1.
35 End Function

```

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**Calculation of HARIN Score:** Algorithm 1 presents our approach for assessing the fit of a topic hierarchy by computing the HARIN score. The `CalcDiversity` function computes the diversity of the child nodes of a given node (Lines 1-10). It initializes the diversity score (Line 2) and assigns a list of the given node's child nodes to `child_list` (Line 3). This process only considers the top- $k$  words of each node. The function starts a loop that iterates over the length of `child_list`. The loop calculates the diversity (Equation 3) using the combinations of child nodes, adding the diversity score (Lines 4-8). The calculated diversity score is divided by the number of combinations of child nodes to derive the average value (Line 9). The function returns the diversity score (Line 10). The `CalcSimilarity` function initializes the similarity score (Line 13) and iterates over each child node, adding the similarity score to their top- $k$  words (Lines 14-16). The average similarity score is derived by dividing the sum of similarity scores by the number of child

**Algorithm 2:** Selection of the Best-fitting Model

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**Input** : A given raw dataset ( $RD$ ), a list ( $L$ ) of candidate hierarchical topic models to be trained using  $RD$ , and the number ( $k$ ) of top words to consider in a topic

**Output**: The selected hierarchical topic model ( $best\_mdl$ )

```

1  $PD \leftarrow Preprocessing(RD)$ ;
2  $best\_mdl \leftarrow None$ ;
3  $best\_score \leftarrow -1$ ;
4 foreach  $m \in L$  do
5    $tm \leftarrow m.train(PD)$ ;
6    $score \leftarrow CalcTotalScore(tm, k)$ ; // See Algo. 1.
7   if  $best\_score < score$  then
8      $best\_mdl \leftarrow tm$ ;
9      $best\_score \leftarrow score$ ;
10  end
11 end
12 return  $best\_mdl$ ;

```

---

nodes (Line 17), and the function returns the calculated similarity score (Line 18). `CalcTotalScore()` produces the final HARIN score on a given hierarchy (Lines 20-35). It takes three arguments: a hierarchical topic model ( $m$ ), the top word counts in each topic ( $k$ ), and a weight vector ( $W$ ) (Line 20). In  $W$ , the three elements— $w_s$ ,  $w_d$ , and  $w_c$ —indicate pre-defined weights for similarity, diversity, and coherence metrics, respectively. We first obtain a topic hierarchy ( $H$ ) produced by  $m$  (Line 21). The function then initializes the diversity and similarity scores (Lines 22-23) and gets the coherence score of the hierarchy (Line 24). Then, we iterate through all nodes in the hierarchy (Lines 25-31). The diversity and similarity scores are averaged (Lines 32-33). Lastly, we calculate and return the final score by applying Equation 1 (Line 34).

## 4.2 Optimal Hierarchical Topic Model Selection by the Hierarchical Harmony Index

Algorithm 2 describes the procedure for selecting the most suitable hierarchical topic model among candidates, using the proposed HARIN metric. The algorithm begins by preprocessing the collected data in the given dataset ( $D$ ) (Line 1). (This step can be skipped if the dataset's preprocessing is unnecessary.) The variables  $best\_mdl$  and  $best\_score$  are initialized (Lines 2-3). For each hierarchical topic model ( $m$ ) in the list ( $L$ ), we train  $m$  using the preprocessed dataset ( $PD$ ) and obtain the trained model ( $tm$ ) (Lines 4-5). We then calculate the HARIN score on  $tm$  with the top  $k$  words in each topic through `CalcTotalScore()`. If the calculated HARIN score is higher than the current best HARIN score ( $best\_score$ ), the  $best\_mdl$  variable is updated with the current model ( $tm$ ), and  $best\_score$  is updated by the current HARIN score ( $score$ ) (Lines 7-10). If the calculated HARIN score is higher than the current best HARIN score ( $best\_score$ ), the  $best\_mdl$  variable is updated with the current model ( $m$ ), and  $best\_score$  is updated with the current HARIN score ( $score$ ) (Lines 7-9). Lastly, the algorithm returns the best model ( $best\_mdl$ ) based on the highest HARIN score (Line 12). This process ensures the identification of the most appropriate hierarchical topic model ( $best\_mdl$ ) for the given dataset ( $D$ ).

## 5 Experiments

**Research Questions.** We designed our experiments to respond to the following research questions.

- **RQ1.** How *consistently* does HARIN choose an optimal hierarchical topic model across various real-world datasets? (Answer: See Table 2)
- **RQ2.** How *similarly* does HARIN rank competing hierarchical topic models as the human evaluators do, compared to other baseline measures? (Answer: See Tables 2 and 3.)
- **RQ3.** How *significantly* does each component of HARIN contribute to the performance? (Answer: See Table 4.)
- **RQ4.** How *effectively* does HARIN prune leaf topics and thus enhance interpretability? (Answer: See Figures 1~3.)

**Environment Settings.** Our experiments were conducted on an Ubuntu 18.04.5 LTS server with the following specifications: Intel Xeon E5-2698 4, Tesla V100-DGXS-32GB, 256 GB RAM, and 7.2 TB SSD. Our algorithms were written in Python version 3.8.19. We also used Konlpy 0.6.0, Tomotopy 0.12, Gensim 4.1.2, and BERTopic 0.14.1. Our source code and dataset are available at [https://github.com/lab-paper-code/covaxx\\_html](https://github.com/lab-paper-code/covaxx_html).

**Considered Models.** Based on code availability, popularity, and representativeness of each category mentioned in Section 2, we used four hierarchical topic models as follows:

- **BERTopic [8]:** This scheme performs sentence embedding using SBERT [33], embedding dimensionality reduction using UMAP [23], clustering using HDBSCAN [31], and extracting topics based on c-TF-IDF [24] from clusters. Through HDBSCAN, BERTopic automatically creates topics without requiring the setting of the number of topics in advance. BERTopic creates hierarchy through bottom-up agglomerative clustering based on the distance between generated clusters. After the hierarchy generation, we set the depth to 3 for a fair comparison with the other models.
- **CluHTM [38]:** This is a hierarchical topic model that utilizes stability analysis metrics to determine the number of topics and captures local information through NMF as well as global semantic information using CluWords [37], a document representation for non-probabilistic matrix factorization based on pre-trained word embeddings to capture global semantic information. The pretrained FastText embedding model (wiki-news-300d-1M) was used for public English datasets. We created a Word2Vec model for collected Korean datasets to use Cluwords.
- **Hierarchical LDA [7] (hLDA):** This model generates a hierarchy based on the concept of nested Chinese Restaurant Process (nCRP) [7]. We trained the hLDA model, which is available in the Python tomatopy package, with 11000 iterations and depth 3. The number of topics automatically derived by hLDA was more than 250, which posed overwhelming challenges in interpretation, as pointed out in Section 1.
- **HyHTM [35]:** This model performs hierarchical topic modeling by incorporating hierarchical information from hyperbolic geometry. We set the depth to 3, as in the other compared models. We applied "poincare\_glove\_50x2D\_cosh-dist-sq\_init\_trick" embedding model and set the hyperparameters as they

**Table 1: Descriptive Statistics for Datasets**

| Datasets           | Description                                                                             | # of records | # of words |
|--------------------|-----------------------------------------------------------------------------------------|--------------|------------|
| COVID-19 (ours)    | Korean tweets mentioning COVID-19 vaccine brands collected from February 2021~July 2021 | 29,297       | 169,031    |
| 20 Newsgroups [34] | Newsgroups posts on 20 topics                                                           | 18,846       | 278,723    |
| PubMed [11]        | Abstract texts from the PubMed papers regarding Type 2 Diabetes Mellitus                | 9,211        | 1,065,818  |
| Yelp [41]          | Yelp text dataset about Korean food discourse                                           | 28,516       | 1,278,126  |
| arXiv [16]         | Abstract texts from the arXiv papers regarding Computer Science                         | 7,945        | 766,232    |

were in the original paper [35]. We used this model only for English datasets because a pre-trained hyperbolic word embedding model for the Korean language was absent.

**Baseline Metrics.** To evaluate the performance of HARIN, we employ several baseline metrics, including topic similarity (Equation 2), topic diversity (Equation 3), and topic coherence (Equation 5). For our ablation studies, we consider all possible combinations of these metrics to justify why HARIN integrates them into a single, unified measure. We compare the performance of these baseline metrics and their combinations against human ratings, in addition to HARIN. We provide further details on how we obtained the human ratings.

**Datasets:** Table 1 exhibits the descriptive statistics of the datasets used in this paper.

- **COVID-19:** This dataset, which we constructed ourselves, includes the tweet discourse of Korean people mentioning COVID-19 vaccine brands during the pandemic. To build this dataset, we collected Korean tweets mentioning at least one keyword, such as 'Novavax,' 'Moderna,' 'AstraZeneca,' 'Janssen,' and 'Pfizer,' via the Twitter API. We set our collection period from February 2021 when the COVID-19 vaccination rollout plan was announced and started, to July 2021 in Korea. The total number of tweets (records) and words obtained after our preprocessing is provided in the first row of Table 1. The preprocessing code of the raw Korean tweets is available on the GitHub repository.
- **20 Newsgroups [34]:** This is a well-known dataset used for text classification and clustering experiments. The dataset has 18,846 newsgroup posts on 20 different newsgroups. As a preprocessing step, we discarded numeric tokens and punctuation, tokenized texts using PorterStemmer, and removed stopwords.
- **PubMed [11]:** This dataset consists of the PubMed abstracts and titles concerning Type 2 Diabetes Mellitus. For evaluation, we used the portion of the abstracts in this dataset. The total number of abstracts is 9,211, and the total number of words in the abstracts was 1,065,818. We applied the same processing as that of the 20 Newsgroups dataset.

- **Yelp** [41]: This dataset is a subset of businesses, reviews, and user data for use in connection with academic research. We extracted and utilized the portion of this dataset related to Korean food. The total number of records was 28,516, and the total number of words was 1,278,126. We applied the same preprocessing as that of the 20 Newsgroups dataset.
- **arXiv** [16]: This dataset consists of the arXiv paper abstracts, titles, and terms about Computer Science. We used a portion of the abstracts about Machine Learning and AI in this dataset. The total abstracts were 7,945, and the total words in the abstracts were 766,232. We applied the same preprocessing as that of the 20 Newsgroups and Yelp datasets as well.

**Evaluation Methodology.** We trained a hierarchical topic model and obtained its topic hierarchy for each dataset. We then calculated the HARIN score on the obtained hierarchy along with Algorithm 1. We applied this calculation to all the combinations of the above four models and five different datasets.

We recruited ten computer science researchers to conduct the human evaluation. The evaluators were asked to assess the quality of the hierarchy based on four criteria:

- (1) “Do you think that a word included in each topic is correctly assigned to that topic?”
- (2) “Do you think each child topic is similar to its parent topic?”
- (3) “Do you think the child topics under the same parent are diverse?”
- (4) “Do you think the parent and child topic words are semantically coherent?”

Each evaluator spent approximately an hour completing the questionnaire, as they had to respond to every question for all model and dataset combinations. These questions were crafted to capture human perceptions of intra- and inter-topic relationships. Specifically, the first question checks whether words are appropriately assigned to topics. The second assesses word-embedding-based similarity in vertical topic relationships, one of the HARIN components. The third evaluates topic diversity, another HARIN component. Lastly, the fourth question examines how often parent and child topic words co-occur, assessing probabilistic semantic coherence, the final component of HARIN.

The questions offered three possible answers: ‘*Absolutely not*’, ‘*Partially*’, and ‘*Yes*’, which we converted to 0, 1, and 2, respectively, based on Nikolenko et al.’s work [27]. We then summed the evaluators’ scores for each of the four questions and calculated the models’ ranking based on the human ratings. To compare the HARIN scores with the evaluators’ ratings, we also ranked the produced hierarchical topic models in the order of each metric’s score. Following this (especially to address RQ2), we calculated the degree of consistency between human ratings and competing measures, as employed in previous work [27].

(The original questionnaires and the evaluators’ responses are available on the GitHub repository as well.)

**Result Analysis I - Optimal Model Choice Consistency (RQ1).** Table 2 addresses the first research question. We generated topic hierarchies from four models for each dataset and evaluated their quality using three baselines (topic similarity, diversity, and coherence), their combinations, the proposed HARIN metric, and

human scores. We then examined whether the model with the highest HARIN score is also optimal across datasets, where ‘optimal’ refers to the model with the maximum human score. For a fair comparison, leaf topics were pruned based on their HARIN score.

In a nutshell, human evaluators consistently rated hLDA the highest across all datasets, which aligned with HARIN’s results. Below are more detailed analyses of the findings around each dataset.

For the COVID-19 dataset, hLDA outperformed CluHTM and BERTopic in HARIN. CluHTM excelled in topic coherence but was not perceived as superior by human evaluators. BERTopic only performed well in topic similarity. The HyHTM model could not be used due to the lack of a pre-trained hyperbolic word embedding model for Korean. (Future work could address this by creating a model using a large Korean text dataset.)

For the 20 Newsgroups dataset, hLDA outperformed the others based on HARIN and human scores as well. While HyHTM was considered, it lagged behind the other models on these two metrics. BERTopic performed best in topic diversity but scored lower on other metrics, including the human score. CluHTM excelled in similarity but did not perform as well in HARIN or human score. HyHTM did not rank highest in any metric.

For the PubMed dataset, hLDA achieved the highest HARIN score (0.52) among all four models, with human evaluators also awarding it the highest score (0.88). Although BERTopic’s HARIN score was close to hLDA’s, its human score (0.69) was lower. CluHTM underperformed, except in similarity and its combinations. HyHTM showed comparable performance in the similarity and coherence combination but did not surpass the other models in HARIN or human scores.

For the Yelp dataset, hLDA once again scored highest in both human and HARIN metrics, while neither BERTopic nor CluHTM ranked highest in any category. HyHTM performed best in similarity and related combinations but was outperformed by hLDA in other metrics, including the human score.

For the arXiv dataset, hLDA secured the highest HARIN and human scores. BERTopic had the second-highest HARIN score, influenced heavily by its performance in topic diversity. CluHTM excelled in similarity and its combinations but did not outperform other models on other metrics. HyHTM performed the lowest in all metrics except the combination of similarity and coherence.

BERTopic constructs a hierarchy of topics in a bottom-up manner, starting with flat topics that are treated as leaf nodes. However, since c-TF-IDF [24] is employed in this process, words that are not commonly used may still be included in the topics as they ascend the hierarchy. Consequently, many words may appear challenging to interpret or even seem meaningless. This issue contributes to the relatively low coherence scores seen in BERTopic (in Table 2).

Moreover, due to the nature of this BERTopic-based hierarchy, the parent topic often resembles the child topic and contains many documents. This leads to a relatively low similarity score when evaluated using the penalty proposed in this paper.

CluHTM utilizes hierarchical topic modeling, incorporating Non-negative Matrix Factorization (NMF) and CluWords. Cluwords are information from pre-trained word embeddings that create clusters of similar semantic words. Due to its heavy reliance on NMF, CluHTM generates more root topics than other models. These results lead to reduced coherence across the datasets. That said,

**Table 2: Performance comparison of eight evaluation metrics on four models across five different datasets (S: topic similarity, D: topic diversity, C: topic coherence, HRN: proposed metric, HUM: human evaluation)**

| Models   | COVID-19 |      |      |      |      |      |      |      | 20Newsgroups |      |      |      |      |      |      |      | PubMed |      |      |      |      |      |      |      | Yelp |      |      |      |      |      |      |      | arXiv |      |      |      |      |      |      |      |
|----------|----------|------|------|------|------|------|------|------|--------------|------|------|------|------|------|------|------|--------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|------|------|------|------|------|------|------|
|          | S        | D    | C    | S+D  | D+C  | S+C  | HRN  | HUM  | S            | D    | C    | S+D  | D+C  | S+C  | HRN  | HUM  | S      | D    | C    | S+D  | D+C  | S+C  | HRN  | HUM  | S    | D    | C    | S+D  | D+C  | S+C  | HRN  | HUM  | S     | D    | C    | S+D  | D+C  | S+C  | HRN  | HUM  |
| BERTopic | 0.61     | 0.46 | 0.35 | 0.53 | 0.40 | 0.48 | 0.47 | 0.66 | 0.43         | 0.58 | 0.48 | 0.51 | 0.53 | 0.46 | 0.49 | 0.56 | 0.19   | 0.84 | 0.51 | 0.51 | 0.68 | 0.35 | 0.51 | 0.69 | 0.17 | 0.82 | 0.51 | 0.50 | 0.67 | 0.34 | 0.50 | 0.64 | 0.23  | 0.78 | 0.50 | 0.50 | 0.64 | 0.36 | 0.49 | 0.68 |
| CluHTM   | 0.23     | 0.41 | 0.70 | 0.32 | 0.56 | 0.46 | 0.44 | 0.56 | 0.58         | 0.49 | 0.44 | 0.53 | 0.47 | 0.51 | 0.50 | 0.70 | 0.39   | 0.71 | 0.38 | 0.56 | 0.55 | 0.39 | 0.49 | 0.59 | 0.29 | 0.73 | 0.40 | 0.51 | 0.56 | 0.35 | 0.47 | 0.48 | 0.46  | 0.57 | 0.35 | 0.51 | 0.46 | 0.40 | 0.45 | 0.60 |
| hLDA     | 0.54     | 0.58 | 0.53 | 0.56 | 0.56 | 0.54 | 0.55 | 0.79 | 0.55         | 0.51 | 0.55 | 0.53 | 0.53 | 0.56 | 0.53 | 0.81 | 0.2    | 0.83 | 0.53 | 0.52 | 0.68 | 0.37 | 0.52 | 0.88 | 0.17 | 0.87 | 0.52 | 0.52 | 0.69 | 0.34 | 0.51 | 0.79 | 0.21  | 0.77 | 0.51 | 0.49 | 0.64 | 0.36 | 0.50 | 0.73 |
| HyHTM    | N/A      |      |      |      |      |      |      |      | 0.52         | 0.47 | 0.40 | 0.49 | 0.43 | 0.46 | 0.46 | 0.53 | 0.41   | 0.57 | 0.37 | 0.49 | 0.47 | 0.39 | 0.44 | 0.45 | 0.35 | 0.81 | 0.37 | 0.58 | 0.59 | 0.37 | 0.51 | 0.36 | 0.44  | 0.52 | 0.35 | 0.48 | 0.44 | 0.40 | 0.43 | 0.43 |

by combining word embedding models with co-occurrence data, CluHTM enhances the similarity between parent and child topics, resulting in a higher overall similarity score.

HyHTM records the lowest HARIN score across key metrics, largely due to the embedding model used ‘poincare\_glove\_50x2D\_cosh-dist-sq\_init\_trick’ which is not well-suited to our dataset. In detail, HyHTM creates diverse words and irrelevant vocabularies. Consequently, HyHTM produces words that are difficult to interpret and form loosely related topics.

In contrast, hLDA achieves the highest HARIN score due to balanced scores across key metrics: the similarity between parent and child topics, diversity among sibling topics, and coherence across topics. Indeed, although hLDA tended to produce a hierarchy of topics based on the probability of words’ appearance, the generated hierarchy fairly preserved the semantics of the words. As will be visualized in Figure 3, hLDA applying HARIN produces a well-balanced topic hierarchy without much information loss at leaf levels.

**Summary:** The results in Table 2 affirm the effectiveness of HARIN scoring in selecting the optimal model for interpreting a given dataset while aligning well with the assessments made by human evaluators.

**Result Analysis II - Ranking Alignment (RQ2).** Tables 2 and 3 address the second research question. The upper portion of Table 3 shows the accuracy with which the model rankings, based on each metric, align with those from human evaluators across different datasets. Given an evaluation metric  $e$ , its accuracy is defined as:

$$\text{Accuracy} = \frac{1}{|D|} \sum_{i=1}^D \frac{1}{|M|} \sum_{j=1}^M f(\text{rank}_{ij}^e, \text{rank}_{ij}^H), \quad (6)$$

where  $D$  is the number of datasets,  $M$  is the number of hierarchical topic models,  $\text{rank}_{ij}^e$  and  $\text{rank}_{ij}^H$  indicate ranks of the  $j$ -th model on the  $i$ -th dataset by the given  $e$  metric and human evaluators, respectively. The function  $f(x, y)$  is defined as:

$$f(x, y) = \begin{cases} 1, & \text{if } x = y \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The function checks if the rank  $x$  equals  $y$ ; if so, it returns 1, and otherwise, 0. The maximum value for accuracy is 1, indicating that the given metric ranks the models exactly in the same order as human evaluators.

**Table 3: Ranking accuracy and quality comparison across different metrics relative to human scores for each dataset**

| Dataset / Metrics | S    | D    | C    | S+D  | D+C  | S+C  | HARIN       |
|-------------------|------|------|------|------|------|------|-------------|
| COVID-19          | 0.33 | 1    | 0    | 1    | 0.33 | 1    | 1           |
| 20 Newsgroups     | 0    | 0.25 | 0.5  | 0.5  | 0.25 | 0.25 | 1           |
| PubMed            | 0    | 0.5  | 1    | 0.25 | 1    | 0    | 1           |
| Yelp              | 0    | 0.5  | 0.5  | 0.25 | 0.5  | 0    | 0.25        |
| arXiv             | 0    | 0.5  | 1    | 0.5  | 1    | 0    | 1           |
| <b>Average</b>    | 0.07 | 0.55 | 0.6  | 0.5  | 0.62 | 0.25 | <b>0.85</b> |
| <b>MRR</b>        | 0.37 | 0.67 | 0.87 | 0.48 | 0.87 | 0.58 | <b>1</b>    |

Overall, HARIN’s rankings are, on average, 85% consistent with the human score-based rankings across the datasets, as shown in the yellow cell of Table 3. In Table 2, human evaluators assigned identical ranks to the models as HARIN except for the Yelp dataset. Specifically, HyHTM was ranked fourth by human evaluators but second by HARIN, resulting in a ranking accuracy of 0.25, as indicated by the orange cell. This lower accuracy was primarily due to ranking discrepancies between HARIN and the human ratings on BERTopic, despite the minimal difference in HARIN scores.

In contrast, the ranking accuracy of other metrics lagged behind HARIN. While the coherence measure showed performance comparable to HARIN, it did not surpass HARIN on the 20 Newsgroups and Yelp datasets. On the COVID-19 dataset, the coherence measure was completely inconsistent with the human evaluations. These findings suggest that the coherence metric alone is insufficient for evaluating the quality of topic hierarchies, as humans also consider factors like topic similarity and diversity in their interpretation of a dataset. HARIN addresses these concerns effectively across diverse datasets.

As a complementary comparison metric, we used Mean Reciprocal Rank (MRR) [9], commonly adopted for measuring ranking quality in information retrieval. MRR is defined as:

$$\text{MRR} = \frac{1}{|M|} \sum_{m=1}^{|M|} \frac{1}{\text{rank}_m}, \quad (8)$$

where  $M$  represents the number of hierarchical topic models, and  $\text{rank}_m$  is the rank assigned to the  $m$ -th model. The maximum MRR value is 1, with higher values indicating better performance. Since MRR gives more weight to the rank of the top model, it is a useful metric for assessing how closely each hierarchical topic model evaluation metric mirrors human rankings.

**Table 4: Correlational analysis results**

| Dataset       | Baseline   | HARIN | Human Score |
|---------------|------------|-------|-------------|
| COVID-19      | Similarity | 0.57  | 0.73        |
|               | Diversity  | 0.99  | 0.98        |
|               | Coherence  | -0.21 | -0.42       |
| 20 Newsgroups | Similarity | 0.29  | 0.67        |
|               | Diversity  | 0.27  | -0.17       |
|               | Coherence  | 0.94  | 0.75        |
| PubMed        | Similarity | -0.81 | -0.84       |
|               | Diversity  | 0.97  | 0.89        |
|               | Coherence  | 0.84  | 0.90        |
| Yelp          | Similarity | -0.23 | -0.94       |
|               | Diversity  | 0.91  | 0.69        |
|               | Coherence  | 0.4   | 0.94        |
| arXiv         | Similarity | -0.96 | -0.81       |
|               | Diversity  | 0.99  | 0.90        |
|               | Coherence  | 0.96  | 0.83        |

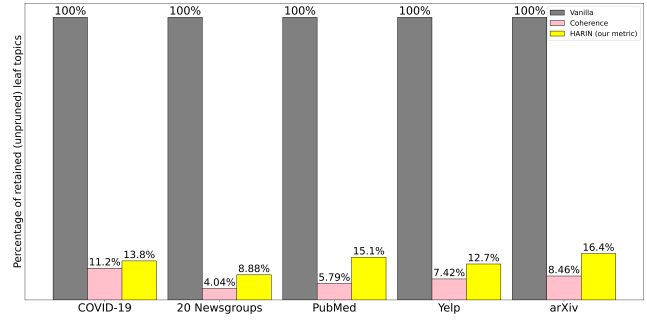
The bottom row of Table 3 shows the MRR evaluation results. HARIN outperforms the compared metrics by achieving an MRR value of 1 for the five disparate datasets. As shown in Table 2, HARIN’s best model selection aligns exactly with that of the recruited evaluators. Although the coherence metric and its combination with the diversity metric are quite competitive, they are still behind HARIN by 13%.

These results suggest that HARIN has a strong potential to replicate human-like ranking performance better than other baseline metrics when selecting models for interpreting a given dataset.

**Result Analysis III - Component Contribution (RQ3).** Table 4 addresses the third research question, presenting the correlation coefficients of similarity, diversity, and coherence with HARIN and human scores across the four models applied to the datasets.

In the COVID-19 dataset, similarity and diversity positively correlated with HARIN, while coherence was negatively correlated. The absolute strength of these correlations followed the order of diversity, similarity, and coherence. These results align closely with human evaluations, suggesting that the models effectively captured various interrelated aspects of COVID-19 discourses, albeit with less emphasis on semantic coherence. Specifically, as shown in Table 2, the diversity (‘D’) and similarity (‘S’) scores were the highest in the selected hLDA model, further affirming the significance of diversity and similarity in relation to HARIN.

In the 20 Newsgroups dataset, all metrics positively correlated with HARIN, whereas diversity negatively correlated with human evaluations. These results suggest that similarity and coherence metrics align well with human judgments, indicating that the models effectively captured coherent and similar topics about 20 news discourses. However, HARIN failed to fully account for the diversity perceived in human scores because HARIN may overlook subtle variations in diversity that humans are more sensitive to. As shown in Table 2, all metrics were well-balanced for the chosen hLDA model.

**Figure 1: Percentage comparison of leaf topics remaining after pruning by each metric: Vanilla vs. Coherence vs. HARIN.**

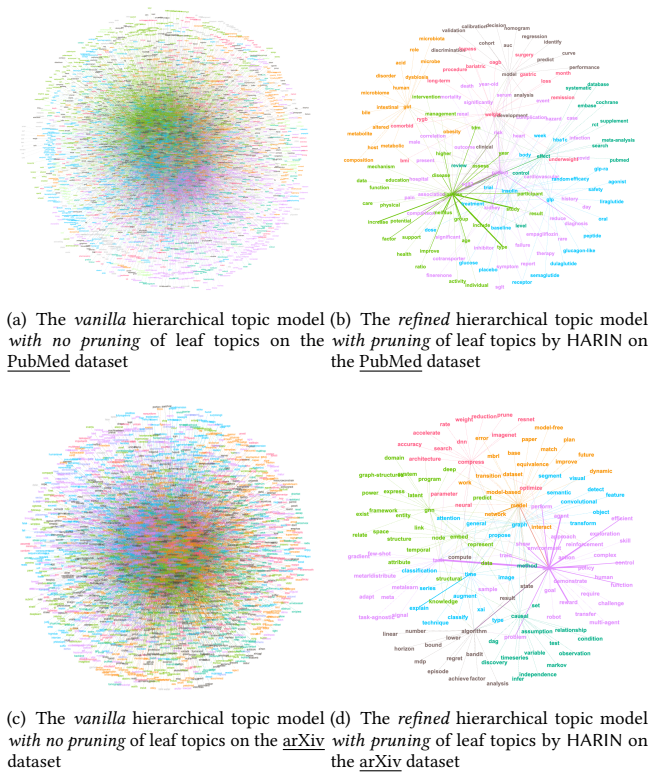
In the PubMed dataset, diversity and coherence showed positive correlations with HARIN and were closely aligned with human evaluations. However, similarity exhibited a negative correlation, indicating struggling to capture relationships between closely related topics. This suggests that the models effectively captured interrelated topics and various aspects of Type 2 Diabetes Mellitus trends also reflected in human evaluations. Notably, as shown in Table 2, the diversity (‘D’) and coherence (‘C’) scores were highest in the selected hLDA model, underscoring the importance of these metrics in relation to HARIN.

In the Yelp dataset, diversity and coherence showed positive correlations with both HARIN and human evaluations, suggesting that the models effectively distinguished and integrated various aspects of discussions on Korean cuisine. However, as in the PubMed dataset, similarity exhibited a negative correlation indicating that models capturing relationships between closely related topics may have presented challenges. Notably, as shown in Table 2, the diversity (‘D’) and coherence (‘C’) scores were highest in the selected hLDA model, reaffirming the importance of these metrics in relation to HARIN.

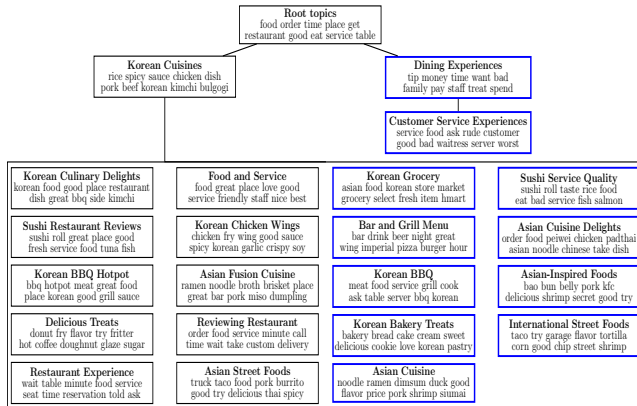
In the arXiv dataset, diversity and coherence exhibited positive correlations with HARIN, while similarity showed a negative correlation. These findings closely align with human evaluations, suggesting that the models coherently captured a broad range of computer science topics. Notably, as shown in Table 2, the diversity (‘D’) and coherence (‘C’) scores were highest in the selected hLDA model, highlighting the critical role these metrics play in capturing the nuances of HARIN.

These findings indicate that the key components contributing to HARIN varied across datasets, yet the overall correlation trends remained consistent with human evaluations. Thus, each component of HARIN plays a significant role in aligning HARIN scores with human assessments.

**Result Analysis IV - Effectiveness of Pruning and Enhancing Interpretability (RQ4).** Figures 1~3 address the fourth research question. Figure 1 illustrates the average percentages of retained leaf topics across the datasets among the four models. The term ‘vanilla’ refers to cases where no leaf topics are pruned, meaning the original leaf topics generated by the hierarchical topic model remain intact. As a result, all vanilla cases show 100% retention



**Figure 2: Visualization result comparison on the PubMed and arXiv datasets: Other datasets were omitted due to space constraints, and all hierarchies have a depth of three.**



**Figure 3: Hierarchy visualization comparison between coherence and HARIN on the Yelp dataset: Blue squares indicate leaf topics added by HARIN.**

in Figure 1. Notably, the coherence metric, our most formidable competitor, prunes on average approximately 5.99% more leaf topics than our HARIN metric. In particular, for the PubMed dataset, the coherence metric pruned about 2.6× more leaf topics than HARIN. This increased pruning by the coherence metric is due to its tendency to decline as the number of topics grows. However, excessive

pruning can lead to significant information loss, making it unsuitable to rely solely on coherence to determine the optimal number of leaf topics. Therefore, HARIN, which prunes fewer topics while maintaining both accuracy and interpretability, is a more favorable alternative. HARIN not only considers coherence but also incorporates similarity and diversity, providing a more comprehensive evaluation of various hierarchical topic models.

Figure 2 compares the visualized topic hierarchies before and after pruning with HARIN. We used Gephi [6] to highlight the differences between the original (vanilla) and retained leaf topics after HARIN pruning. We focus on the hierarchies of the PubMed and arXiv datasets due to space constraints, and these datasets show notable differences. The number of vanilla leaf topics was 1,233 for PubMed and 1,006 for arXiv. Each tree is color-coded to represent distinct topics, with bold lines indicating connections between root and parent topics and thin lines showing connections between parent and child topics. We selected the hLDA model, which was identified as the best-performing model in Table 2. Before pruning, the abundance of leaf topics made interpretation difficult, as shown in Figures 2(a) and 2(c). However, after pruning with HARIN, the average number of leaf topics was reduced to 15, resulting in much clearer and more interpretable visualizations, as seen in Figures 2(b) and 2(d).

Figure 3 compares the hierarchical visualizations of topics pruned by the coherence metric and by HARIN, using the hLDA model and the Yelp dataset due to space constraints. In addition, each topic box displayed the top 10 words. Topics retained by both coherence and HARIN are marked in black, while those retained only by HARIN (but pruned by coherence) are highlighted in blue. The visualization shows that coherence often overprunes, potentially leading to “information loss.” In contrast, HARIN preserves more meaningful topics and provides a more accurate representation. While coherence simplifies categories such as Korean foods, Asian foods, and restaurant experiences, HARIN captures greater diversity by retaining more specific topics related to Korean cuisine. For example, HARIN identifies detailed topics such as Korean grocery stores and bakeries.

These results show that HARIN effectively identifies and retains significant topics that coherence might overlook, resulting in better pruning and enhanced interpretability compared to other baseline metrics.

## 6 Conclusion and Future Work

In this paper, we tackled the challenge of evaluating the interpretability of hierarchical topic models on diverse datasets. Existing metrics expose two major drawbacks: they (1) fail to adequately capture the unique structural characteristics of hierarchical topic models and (2) often over-prune leaf topics, diminishing the interpretability of the resulting models. To address these concerns, we proposed a novel evaluation metric called HARIN, which holistically assesses the quality of hierarchical structures by integrating topic similarity, diversity, and coherence.

We validated HARIN on several real-world datasets, including ours, to determine the most suitable model for each dataset among four representative hierarchical topic modeling approaches:

BERTopic, CluHTM, hLDA, and HyHTM. Our experiments demonstrated that hLDA consistently achieved the highest HARIN scores, aligning closely with human evaluators, who found its hierarchical topic structures more interpretable than competing models. These results confirm HARIN's capability to automatically and effectively identify a given dataset's optimal hierarchical topic model, enhancing interpretability with minimal pruning of meaningful leaf topics. To support reproducibility and further exploration, our code and datasets are openly accessible on GitHub.

Future research can explore several promising directions. First, developing an algorithm to automatically determine the optimal depth of topic hierarchies in top-down methodologies while concurrently optimizing the HARIN score would be a significant advancement. Also, exploring a broader range of models, such as hierarchical topic graphs [40] and large language model-based topic models [30], or devising entirely new approaches, could provide further insights. Refining methods to identify the optimal number of words for each topic also represents an important avenue for improvement. Expanding our analysis to include social media texts from emerging platforms [25] would offer new perspectives and enhance the generalizability of our findings. Finally, we aim to develop techniques for dynamically assigning weights to each component of HARIN, customized to the unique characteristics of individual datasets, thereby improving its flexibility and overall effectiveness.

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